RED EYE FLIGHT (NAPZ) TEAM 11

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PROBLEM OVERVIEW

We decided to focus on the consumer's point of view, specifically answering these questions:

1. "Am I paying above or below the average market fare?"
2. "Which airline would be the best choice in terms of prices?
3. "How confident can I be of this?"

The goal of our analysis is for **individual customers** to be able to predict the average market fare for a flight that they wish to take.

DATA APPROACH

01

FIND RELATIONS IN EDA

Heatmap helped use decide which variables to focus one

02

MORE GRAPHS + FEATURE ENGINEERING

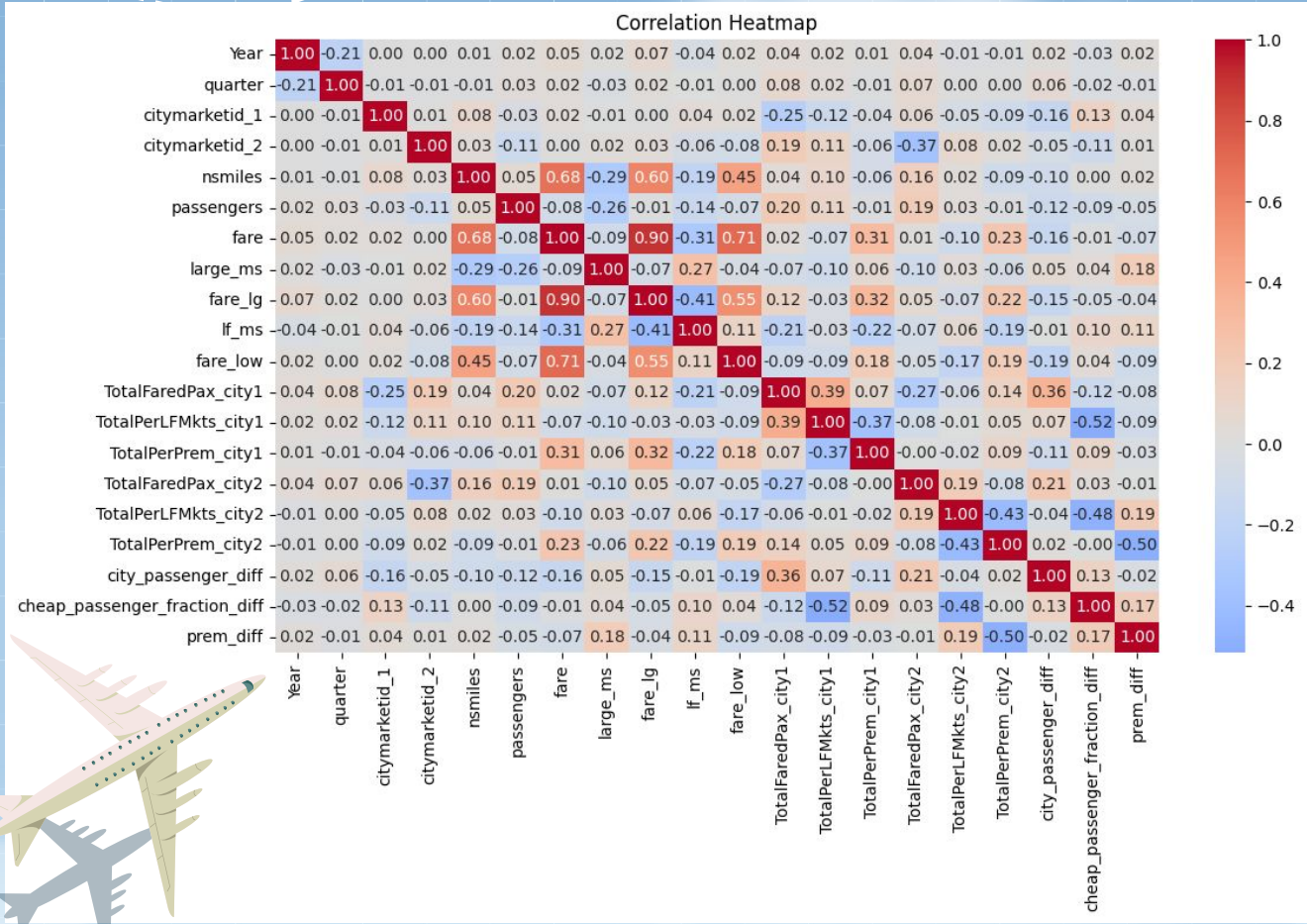
Create more detailed graphs to see the relations, select features

03

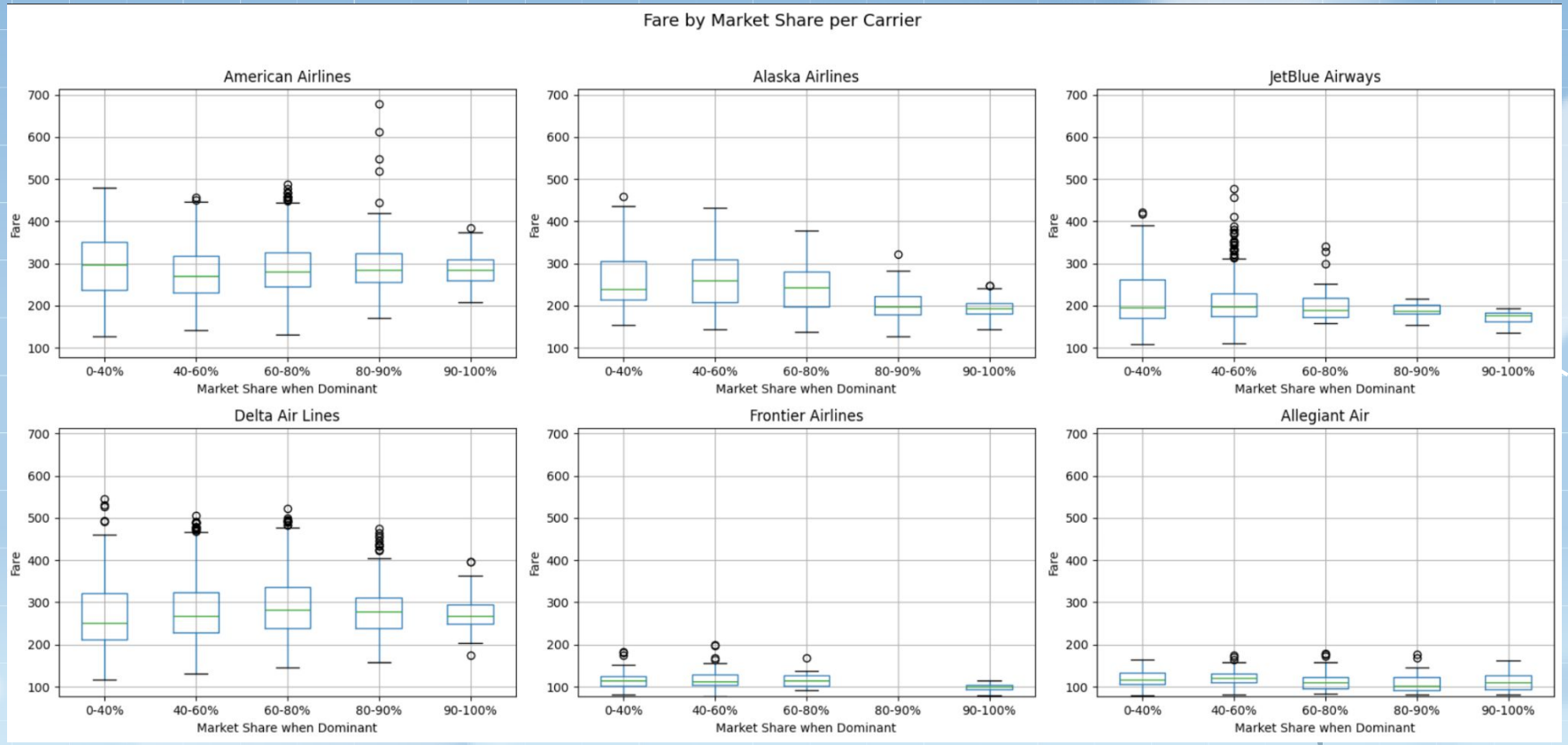
MAKE THE MODEL

Predict and compare prices

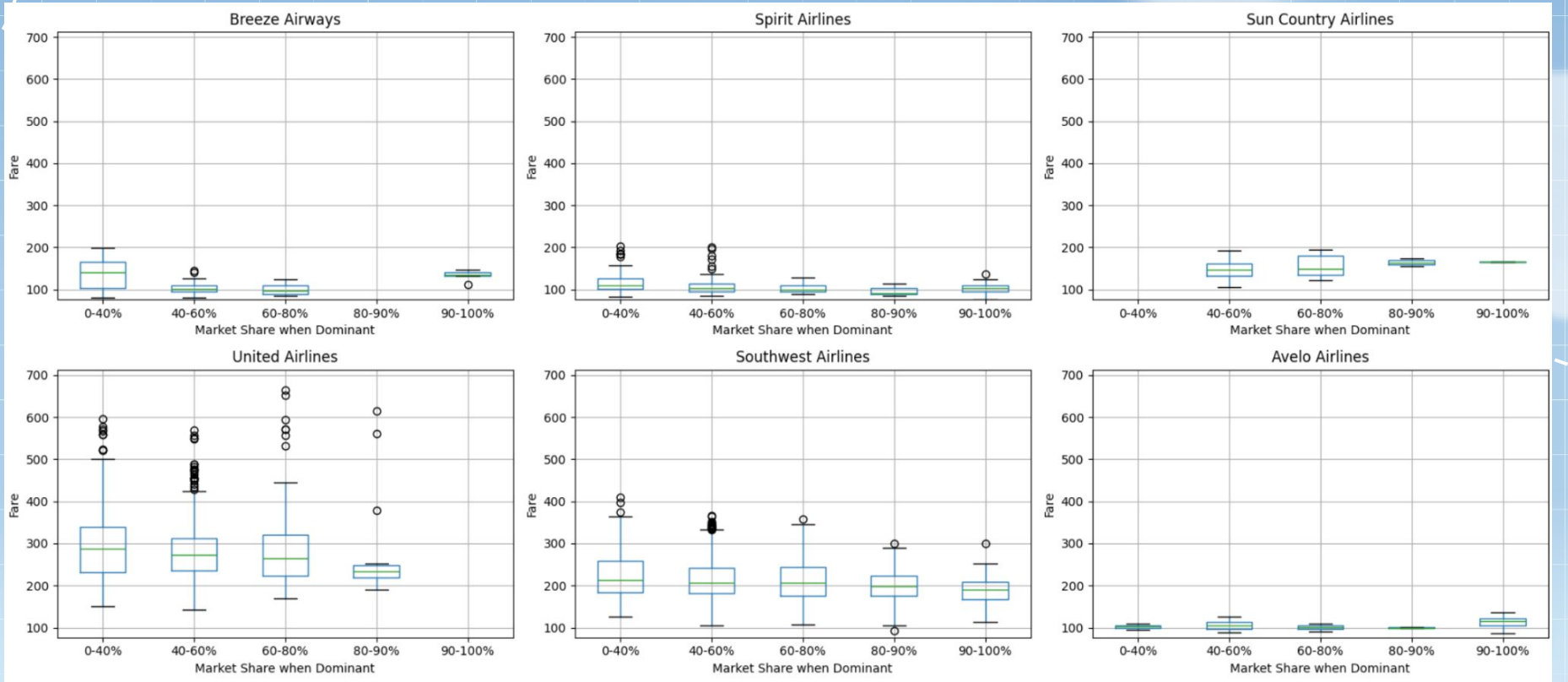
HEAT MAP


Correlation Heatmap
Year 1.00 -0.21 0.00 0.00 0.01 0.02 0.05 0.02 0.07 -0.04 0.02 0.04 0.02 0.01 0.04 -0.01 -0.01 0.02 -0.03 0.02
quarter -0.21 1.00 -0.01 -0.01 -0.01 0.03 0.02 -0.03 0.02 -0.01 0.00 0.08 0.02 -0.01 0.07 0.00 0.00 0.06 -0.02 -0.01
citymarketid_1 0.00 -0.01 1.00 0.01 0.08 -0.03 0.02 -0.01 0.00 0.04 0.02 -0.25 -0.12 -0.04 0.06 -0.05 -0.09 -0.16 0.13 0.04
citymarketid_2 0.00 -0.01 0.01 1.00 0.03 -0.11 0.00 0.02 0.03 -0.06 -0.08 0.19 0.11 -0.06 -0.37 0.08 0.02 -0.05 -0.11 0.01
nsmiles 0.01 -0.01 0.08 0.03 1.00 0.05 0.68 -0.29 0.60 -0.19 0.45 0.04 0.10 -0.06 0.16 0.02 -0.09 -0.10 0.00 0.02
passengers 0.02 0.03 -0.03 -0.11 0.05 1.00 -0.08 -0.26 -0.01 -0.14 -0.07 0.20 0.11 -0.01 0.19 0.03 -0.01 -0.12 -0.09 -0.05
fare 0.05 0.02 0.02 0.00 0.68 -0.08 1.00 -0.09 0.90 -0.31 0.71 0.02 -0.07 0.31 0.01 -0.10 0.23 -0.16 -0.01 -0.07
large_ms 0.02 -0.03 -0.01 0.02 -0.29 -0.26 -0.09 1.00 -0.07 0.27 -0.04 -0.07 -0.10 0.06 -0.10 0.03 -0.06 0.05 0.04 0.18
fare_lg 0.07 0.02 0.00 0.03 0.60 -0.01 0.90 -0.07 1.00 -0.41 0.55 0.12 -0.03 0.32 0.05 -0.07 0.22 -0.15 -0.05 -0.04
lf_ms -0.04 -0.01 0.04 -0.06 -0.19 -0.14 -0.31 0.27 -0.41 1.00 0.11 -0.21 -0.03 -0.22 -0.07 0.06 -0.19 -0.01 0.10 0.11
fare_low 0.02 0.00 0.02 -0.08 0.45 -0.07 0.71 -0.04 0.55 0.11 1.00 -0.09 -0.09 0.18 -0.05 -0.17 0.19 -0.19 0.04 -0.09
TotalFaredPax_city1 0.04 0.08 -0.25 0.19 0.04 0.20 0.02 -0.07 0.12 -0.21 -0.09 1.00 0.39 0.07 -0.27 -0.06 0.14 0.36 -0.12 -0.08
TotalPerLFMkts_city1 0.02 0.02 -0.12 0.11 0.10 0.11 -0.07 -0.10 -0.03 -0.03 -0.09 0.39 1.00 -0.37 -0.08 -0.01 0.05 0.07 -0.52 -0.09
TotalPerPrem_city1 0.01 -0.01 -0.04 -0.06 -0.06 -0.01 0.31 0.06 0.32 -0.22 0.18 0.07 -0.37 1.00 -0.00 -0.02 0.09 -0.11 0.09 -0.03
TotalFaredPax_city2 0.04 0.07 0.06 -0.37 0.16 0.19 0.01 -0.10 0.05 -0.07 -0.05 -0.27 -0.08 -0.00 1.00 0.19 -0.08 0.21 0.03 -0.01
TotalPerLFMkts_city2 -0.01 0.00 -0.05 0.08 0.02 0.03 -0.10 0.03 -0.07 0.06 -0.17 -0.06 -0.01 -0.02 0.19 1.00 -0.43 -0.04 -0.48 0.19
TotalPerPrem_city2 -0.01 0.00 -0.09 0.02 -0.09 -0.01 0.23 -0.06 0.22 -0.19 0.19 0.14 0.05 0.09 -0.08 -0.43 1.00 0.02 -0.00 -0.50
city_passenger_diff 0.02 0.06 -0.16 -0.05 -0.10 -0.12 -0.16 0.05 -0.15 -0.01 -0.19 0.36 0.07 -0.11 0.21 -0.04 0.02 1.00 0.13 -0.02
cheap_passenger_fraction_diff -0.03 -0.02 0.13 -0.11 0.00 -0.09 -0.01 0.04 -0.05 0.10 0.04 -0.12 -0.52 0.09 0.03 -0.48 -0.00 0.13 1.00 0.17
prem_diff 0.02 -0.01 0.04 0.01 0.02 -0.05 -0.07 0.18 -0.04 0.11 -0.09 -0.08 -0.09 -0.03 -0.01 0.19 -0.50 -0.02 0.17 1.00
Year
quarter
citymarketid_1
citymarketid_2
nsmiles
passengers
fare
large_ms
fare_lg
lf_ms
fare_low
TotalFaredPax_city1
TotalPerLFMkts_city1
TotalPerPrem_city1
TotalFaredPax_city2
TotalPerLFMkts_city2
TotalPerPrem_city2
city_passenger_diff
cheap_passenger_fraction_diff
prem_diff
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

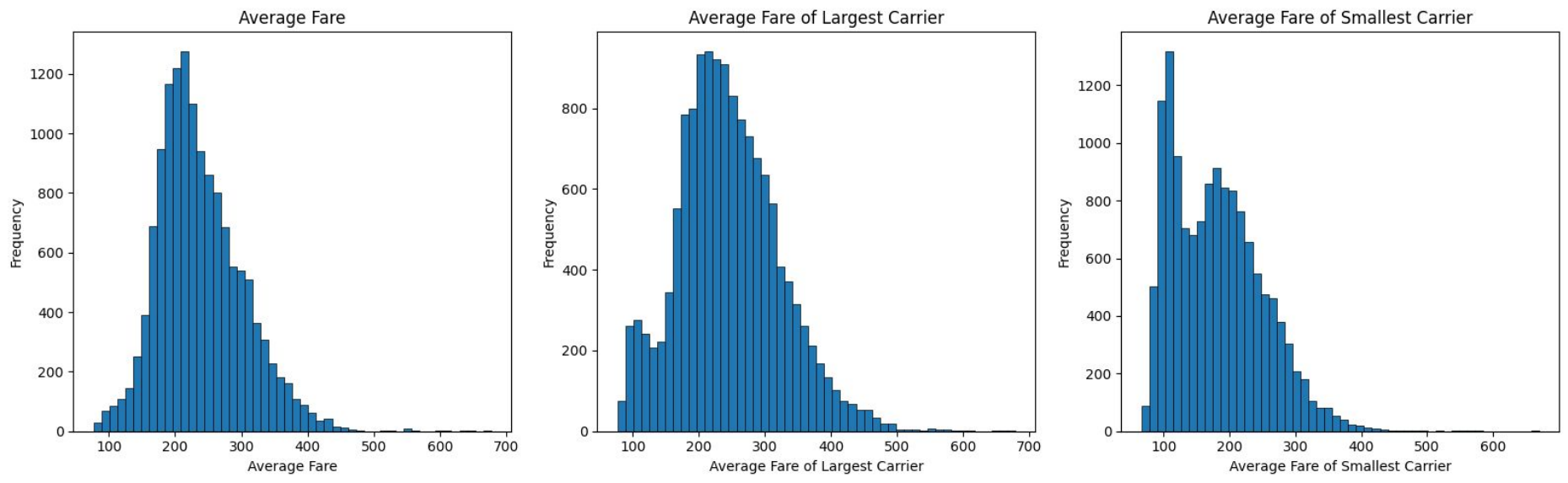
MARKET SHARE TO FARE



MARKET SHARE TO FARE

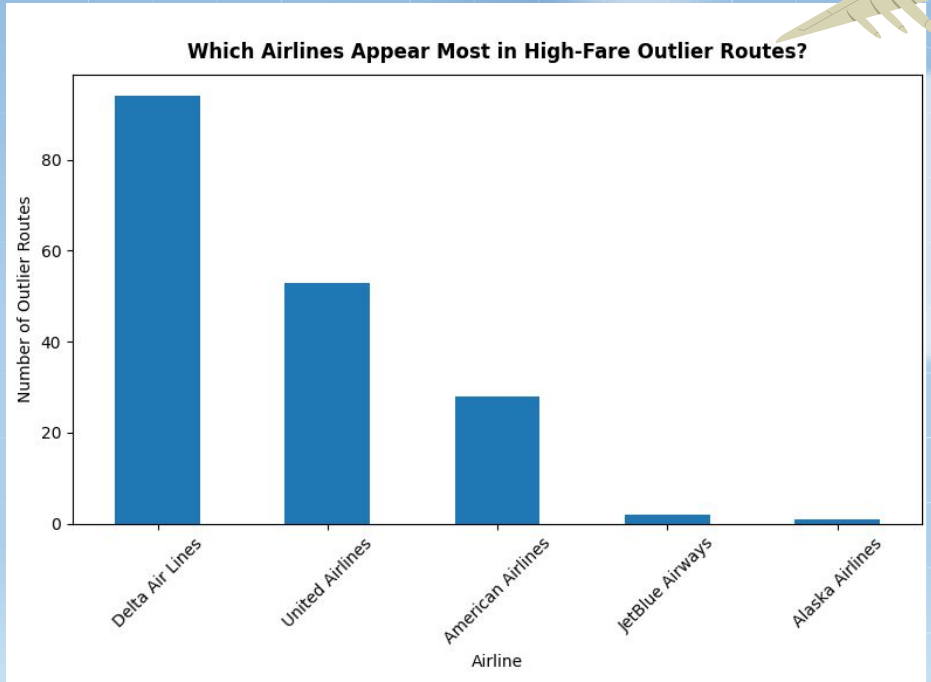


MARKET SHARE TO FARE



OUTLIER ANALYSIS

- Rare cities tend to be outliers in fare, but on average cost less with a smaller spread.
- Notably, Eagle, Aspen, CO, and Jackson are ski destinations (Vail for Eagle), with flights only offered in Q1 (Jan-March)
  - Jackson Hole does offer flights in Q3, however
- Ski trips and other high volume tourist destinations could be the reason for the high fare



FEATURE ENG.

- Cut city1 and city2
- Cut "TotalFaredPax_city1", "TotalFaredPax_city2"
- Quarters since Q1 2021
- Dummy variables
- Log-transform based on skew of target variables

MODEL

Let $X \in \mathbb{R}^{(n \times p)}$ be the scaled feature matrix and $y \in \mathbb{R}^n$ the log-transformed target vector. The model is:

$\alpha \sim \text{Normal}(\bar{y}_\text{train}, 1)$ # intercept, centred on training mean

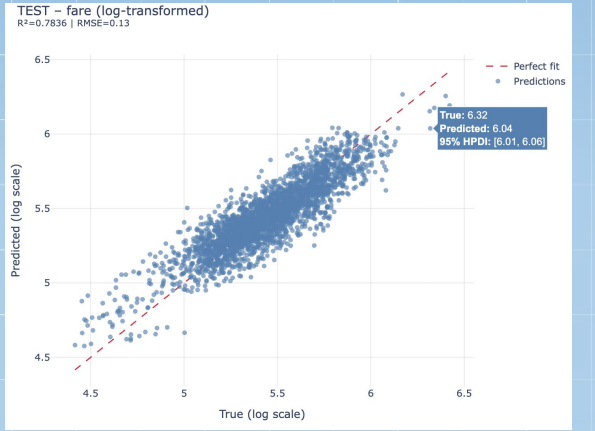
$\beta_j \sim \text{Normal}(0, 0.5)$ for $j=1..p$ # regularizing priors to avoid overfitting

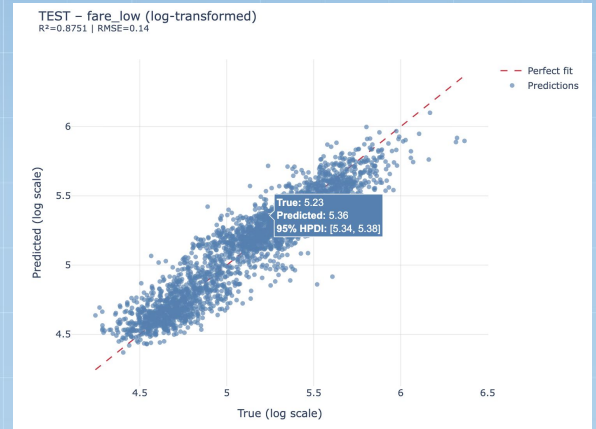
$\sigma \sim \text{HalfNormal}(1)$ # observation noise, constrained > 0

$\mu_i = \alpha + X_i \cdot \beta$ # linear predictor

$y_i \sim \text{Normal}(\mu_i, \sigma)$

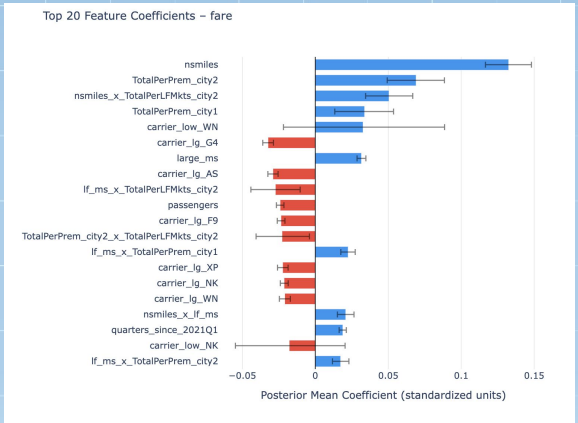
TESTING FIT

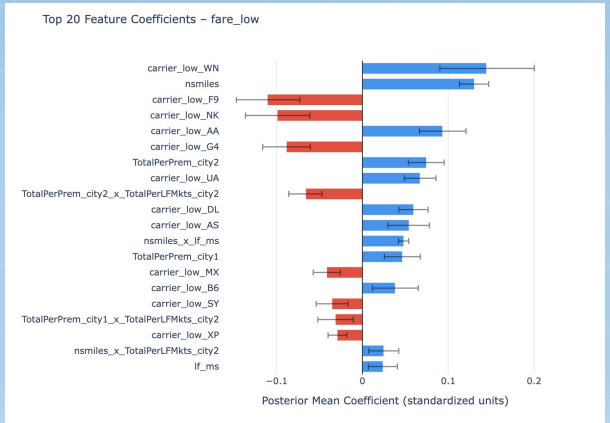

TEST – fare (log-transformed)
R²=0.7836 | RMSE=0.13
Perfect fit
Predictions
True: 6.32
Predicted: 6.04
95% HPDI: [6.01, 6.06]
Predicted (log scale)
True (log scale)
4.5
5
5.5
6
6.5

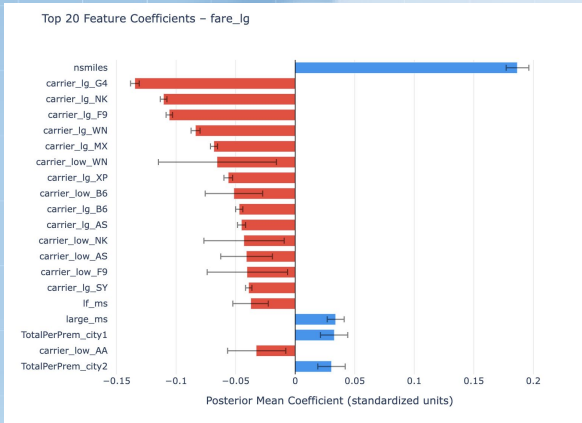

TEST – fare_low (log-transformed)
R²=0.8751 | RMSE=0.14
Perfect fit
Predictions
True: 5.23
Predicted: 5.36
95% HPDI: [5.34, 5.38]
Predicted (log scale)
True (log scale)
4.5
5
5.5
6
6.5

TEST – fare_lg (log-transformed)
R²=0.8522 | RMSE=0.13
Perfect fit
Predictions
True: 5.05
Predicted: 5.54
95% HPDI: [5.51, 5.56]
Predicted (log scale)
True (log scale)
4.5
5
5.5
6
6.5

VARIABLE IMPORTANCE AND UNCERTAINTY







PREDICTION TOOL

Route: citymarketid 32467 → 32457 | Quarter since Q1 2021: 3

Target	Mean ($)	95% CI Lower ($)	95% CI Upper ($)
fare	$351.74	$312.91	$393.39
fare_low	$249.68	$222.26	$278.72
fare_lg	$453.06	$408.55	$500.98

Predictions in original dollar scale (back-transformed from log1p). Credible intervals reflect posterior parameter uncertainty over 4000 MCMC samples.

KEY RESULTS

ON AIRLINES

Airlines' market share affects their pricing, particularly when they are the primary airline on the market.

MODEL

Achieved a well-fitting model that does not overfit and is highly explainable

TOOL

Created a tool for customers to predict the average fare price and make more informed decision

FINAL INSIGHTS

Spend money wisely